
Re: Test Invitaion - Junior Credit Research Analyst

From wwwzzy1@connect.hku.hk <wwwzzy1@connect.hku.hk>

Date Tue 2025-12-09 17:01

To 张凯澄(Erin Cheung) <erin.hc.cheung@pingan.com.hk>

Cc chris.qc.zhou@pingan.com.hk <chris.qc.zhou@pingan.com.hk>; mandy.fung@pingan.com.hk <mandy.fung@pingan.com.hk>

Dear Erin,

Thanks for your email. I have received the test materials and the attached report. I accept the task and will submit the deliverables by **Monday, 15 Dec 2025**.

To ensure my output aligns with your expectations, I have a few quick questions regarding the scope:

1. Methodology & Tuning

The email indicates that the specific parameters (e.g., 3m/6m/1y moments vs. 1y/3y/5y lookbacks) are for illustrative purposes. My understanding is that the **core construction methodology** (aggregating the four moments: Returns, Volatility, Momentum, and Fragility, followed by the logistic scaling) remains the fixed standard, while the **time windows** could be adapted to the specific asset's dynamics.

Therefore, I propose to proceed with the following two-step approach, **if it aligns with your expectations**:

- **Step 1 (Baseline)**: Strictly replicate the BofA BRI logic using the report's default parameters to establish a benchmark.
- **Step 2 (Optimization)**: Conduct sensitivity testing on the lookback windows (e.g., testing shorter windows for high-velocity assets like Crypto) to determine the most effective parameter set for each asset class, as suggested in your notes.

2. Tooling & Format

I plan to use **Python** for the data fetching, modeling, and calculation steps to ensure the model is robust and scalable. I will then synthesize the findings into a report (PDF/PPT). Is this workflow acceptable, or is a functional **Excel based model** explicitly required for this submission?

3. Asset Universe

The report focuses heavily on Equities (Mag7, Nasdaq, HSCEI), Commodities (Gold, Oil), and Bitcoin. I will cover these to validate the model. However, are there other assets you are particularly interested in, such as major China indices (e.g., CSI 1000)?

Additionally, regarding the final output, would you prefer I present:

- **A single asset** worked through in significant depth (showing full intermediate calculation steps and historical backtesting results), or
- **A representative set of assets** (e.g., across Equities, Crypto, and Commodities) to demonstrate the model's versatility?

4. Data Sources

Does the team have a preferred data source for this exercise, or should I source the historical price data myself?

Thank you for your guidance. I look forward to working on this challenge.

Best regards,

Zeyu ZHOU

From: 张凯澄(Erin Cheung) <erin.hc.cheung@pingan.com.hk>

Sent: Tuesday, December 9, 2025 11:31

Cc: 冯敏贤(Mandy Fung) <mandy.fung@pingan.com.hk>

Subject: Test Invitaion - Junior Credit Research Analyst

This is an external email.

Dear Candidate,

Thank you for your interest in our Junior Credit Research Analyst position, and for your time to participate in this test.

The BRI is constructed as an average of 3 sub-indicators, which each range between 0 and 1: a short-term, medium-term, and long-term sub-indicator. The aggregation of these different indicators helps to capture the asset's dynamics on numerous time horizons, giving a more complete assessment of the price action's bubble-like dynamics. Higher values of the BRI would suggest more bubble-like price behaviour.

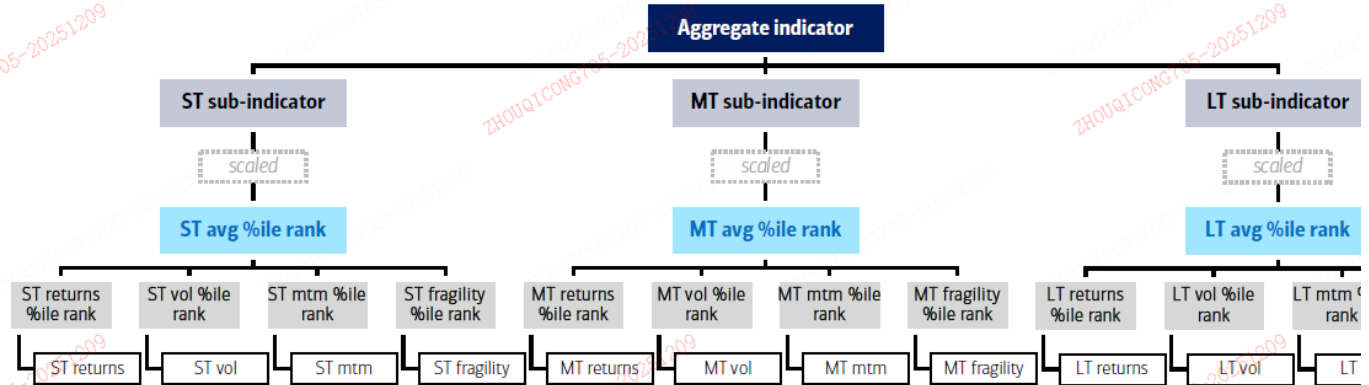
Each sub-indicator aims to capture how stretched the asset's four moments (returns, vol, momentum, and fragility) are relative to its history, and is computed as the average of each of these four moments' percentile rank vs history.⁹ However, given detection of bubble-like behaviour is the primary aim of the indicator, we condition this average percentile rank on periods when the asset is rallying by scaling it up or down depending on how positive or negative returns are.¹⁰ Exhibit 156 summarizes the construction of the aggregated indicator and its microstructure.

This report uses the following parameters for the 3 sub-indicators: (i) short-term = 3m moments vs 1y percentile rank lookback, (ii) medium-term = 6m moments vs 3y percentile rank lookback, (iii) long-term = 1y moments vs 5y percentile rank lookback. These parameter choices are for illustrative purposes and may differ in their efficacy depending on

the dynamics of the asset being analyzed.

10 Scaling factor ranges between 0 & 1. It is computed using a logistic function transformation of the trailing return (tenor based on sub-indicator). Highly positive returns give a factor of ~1 and highly negative returns give a factor of ~0.

Exhibit 156: The aggregated Indicator is an average of 3 sub-Indicators which capture short-, medium- and long-term price dynamics. Each computed as a scaled average percentile rank of the four metrics (returns, vol, momentum & fragility)⁹
Construction summary of BofA Bubble Risk Indicator



Source: BofA Global Research. ST, MT & LT refer to short-, medium- & long-term; 'mtm' refers to momentum component; '%ile rank' refers to percentile rank.

Please find attached material for your reference. Based on this process, please build a similar bubble risk indicator.

Kindly email back the deliverables by 15 Dec 2025 (Monday). Please let us know if you need clarification on the requests. Thank you!

Regards,
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